Time-Arbitrage Scheduling for Heterogeneous Cloud Computing: Learning from Power Grid Dispatch

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Abstract

Cloud computing resource scheduling faces challenges strikingly similar to power grid dispatch: both must balance supply and demand across time with heterogeneous resources while minimizing costs and meeting quality-of-service guarantees. While power systems have developed sophisticated time-shift mechanisms (pumped storage, batteries) over decades, cloud scheduling remains predominantly space-focused. In this paper, we propose a time-arbitrage scheduling framework that explicitly models temporal dimensions in resource allocation. Our approach includes: (1) a multi-level temporal hierarchy (seconds to days), (2) a cost function incorporating time-shifted pricing, and (3) a deadline-aware scheduler that defers non-urgent tasks to low-price periods. Evaluated on synthetic workloads mimicking AI agent platforms, our method reduces costs by 92.8% while maintaining 100% task completion rate, compared to round-robin baselines. This work establishes the first formal analogy between power grid dispatch and cloud scheduling, opening new research directions in temporal resource optimization.

Keywords: Cloud Computing, Resource Scheduling, Time-Arbitrage, Power Grid Analogy, Cost Optimization, Heterogeneous Resources

1 Introduction

Cloud computing has become the backbone of modern digital infrastructure, supporting everything from web services to AI workloads. However, resource utilization remains notoriously inefficient: data centers typically operate at 30-40% average utilization, with peak-to-valley ratios reaching 3-5× [1]. This inefficiency translates to massive economic waste—global cloud spending exceeded $500 billion in 2025, with an estimated $200 billion wasted on idle or underutilized resources [2].

The root cause lies in the temporal mismatch between resource supply and demand. User requests arrive in bursts (business hours, product launches, viral events), while resources must be provisioned for peak capacity. Traditional schedulers focus on *spatial* optimization—placing tasks on available servers—but largely ignore *temporal* optimization: shifting tasks across time to exploit price variations.

1.1 The Power Grid Analogy

Interestingly, power systems face an identical challenge: electricity demand fluctuates throughout the day, while generation capacity must be balanced in real-time. Over decades, power grids have developed sophisticated time-shift mechanisms:

- **Pumped hydro storage**: Pump water uphill during low-demand periods, release through turbines during peaks
- **Battery storage**: Store excess solar/wind energy, discharge when needed
- **Time-of-use pricing**: Incentivize consumers to shift consumption to off-peak hours

1.2 Research Question

Can we apply similar time-arbitrage principles to cloud resource scheduling?

1.3 Our Approach

We propose a time-arbitrage scheduler that:

1. Identifies deferrable tasks (batch jobs, training workloads, non-urgent inference)
2. Delays execution during high-price periods (business hours)
3. Concentrates execution during low-price periods (nights, weekends)
4. Ensures deadlines are met through urgency-aware prioritization

1.4 Contributions

This paper makes three contributions:

1. **Theoretical framework**: First formal model of time-arbitrage in cloud scheduling, including cost functions and optimization objectives (§3)
2. **Algorithm design**: Practical deadline-aware scheduler with provable properties (§4)
3. **Empirical validation**: Comprehensive evaluation showing 92.8% cost reduction with 100% task completion (§5)

2 Related Work

2.1 Cloud Resource Scheduling

Cloud scheduling has been extensively studied, with production systems like Kubernetes [4], Borg [5], and Omega [6] handling millions of tasks daily. These systems focus on *spatial* optimization: bin-packing tasks onto servers to maximize utilization while respecting constraints.

2.2 Time-Aware Scheduling

Recent works have begun exploring temporal dimensions: TempoScale [9] predicts cloud workloads using multi-scale time features. PRISM [10] forecasts GPU cluster workloads. QTIS [11] applies quantum optimization to time-interval scheduling.

2.3 Cost Optimization

Cost-aware scheduling has gained traction with the rise of spot instances: LeJOT [12] optimizes job costs on Databricks. Ksurf-Drone [13] uses contextual bandits for cloud resource allocation.

3 System Model

3.1 Resource Model

We model a heterogeneous resource pool: $R = \{r_1, r_2, ..., r_n\}$. Each resource has type (CPU/GPU/NPU/MEM), capacity, time-varying price $p_i(t)$, and warm-up time.

3.2 Task Model

Tasks arrive over time: $J = \{j_1, j_2, ..., j_m\}$. Each task has arrival time, resource demand, estimated duration, deadline, and deferrability $\delta \in [0,1]$.

3.3 Cost Model

Total Cost = C_resource + C_migration + C_delay + C_violation

4 Time-Arbitrage Scheduler Design

4.1 Urgency Assessment

For each task, compute time to deadline and classify: Critical (<2 min), Urgent (<10 min), Soon (<30 min), Normal (≥30 min).

4.2 Price-Level Detection

Classify current time: High (10-16, 20-23), Medium (6-9, 17-19), Low (otherwise).

5 Evaluation

5.1 Results

Cost: Round-Robin $3.59 → Time-Arbitrage $0.26 (**92.8% savings**)

Completion: 100% maintained

Latency: 156s vs 156s (no change)

6 Conclusion

We presented the first time-arbitrage scheduler for heterogeneous cloud computing, inspired by power grid dispatch. By deliberately deferring non-urgent tasks to low-price periods, we achieve **92.8% cost reduction** while maintaining **100% task completion**.

References

[1] Barroso et al. The datacenter as a computer, 2013.

[2] Gartner. Cloud Computing Spending Forecast, 2025.

[3] Denholm et al. The value of energy storage for grid applications, 2016.

[4-25] See paper_draft_v1.md for full references.

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